

TEMPERATURE, PERFUSION, AND PHYSIOLOGIC CONTEXT SENSING

Physiologic Context Sensing Overview

In addition to mechanical rigidity sensing, the wearable device may incorporate physiologic context sensing modalities configured to improve interpretation of erection state, reduce ambiguity caused by environmental conditions, and enable richer longitudinal physiologic profiling. Context sensing may include temperature measurement, optical perfusion sensing, impedance-based vascular proxies, or other biometric indicators that correlate with vascular activity and autonomic response patterns.

Temperature Sensing Embodiments

One or more temperature sensing elements may be incorporated to measure tissue-adjacent temperature, device internal temperature, and/or ambient reference temperature. Temperature measurements may provide physiologic context including thermal trends associated with vascular activity, arousal episodes, sleep state transitions, or recovery dynamics. Temperature may further be used for environmental compensation and drift correction of strain, pressure, piezoelectric, or magnetic sensing elements whose outputs may vary with temperature.

Temperature sensors may include thermistors, digital temperature sensors, resistive temperature devices, infrared sensing elements, distributed temperature arrays, or other thermal measurement technologies. Sensors may be positioned on an inner surface adjacent tissue, embedded within the wearable body, located near electronics for calibration reference, or placed externally to provide ambient context.

Optical Perfusion and Pulse Sensing Embodiments

Optional embodiments may incorporate photoplethysmography sensing configured to measure pulse waveforms, pulse rate, pulse amplitude proxies, vascular response dynamics, perfusion-related indices, or other optically derived blood-flow-correlated signals. Optical sensing may be implemented using one or more light emitters and one or more detectors configured to measure reflected or transmitted light through tissue or adjacent structures. Sensing may employ single-wavelength or multi-wavelength operation and may include multi-emitter or multi-detector arrangements.

Optical sensing components may be positioned to optimize signal quality, reduce motion sensitivity, and minimize optical leakage through shielding, mechanical isolation, surface texturing, or conformal contact structures. Optical sampling may be continuous, periodic, burst-based, event-triggered, or combined with motion-aware gating determined by inertial sensing outputs.

Additional Context Sensing Modalities

Additional physiologic context sensing may include electrical impedance sensing, bioimpedance-based tissue characterization, dielectric sensing, capacitive proximity sensing, acoustic sensing, or other modalities capable of producing vascular or tissue-condition proxies. Such context channels may be used for enhanced confidence scoring, episode segmentation, artifact rejection, and individualized baseline normalization.

Context-Aware Interpretation

Context sensing outputs may be computationally combined with rigidity sensing modalities to derive composite indices representing erection quality, physiologic readiness, recovery behavior, or longitudinal variability. The disclosed embodiments are illustrative and non-limiting and encompass future biometric context sensing technologies capable of improving longitudinal penile physiology characterization.